

3.1

(1). 最大饱和不失真幅度 $V_{om1} = V_{CC} - V_{CES}$
 $= 12V$

$V_{om} = 12V$

$P_{om} = \frac{V_{om}^2}{2R_L} = \frac{12^2}{2 \times 16} = 4.5W$

(2). 电源功率: $P_E = \frac{2V_{CC} V_{om}}{\pi R_L} = \frac{2 \times 12 \times 12}{\pi \times 16}$
 $= 5.730W$

管耗总功率 $P_T = P_E - P_{om} = 1.230W$

$P_{CM} \geq \frac{1}{2} P_T = 0.615W$ $\frac{|V_{o1}|_{max} + V_{CC}}{2}$

(3). $|V_{CEQ}| > |V_{CE}|_{max} = 2|V_{om}|_{max} = 2V_{om} = 24V$

3.2 最大饱和不失真 $V_{om1} = |V_{CC} - V_{CES}| = |V_{CC}|$

$V_{om} = V_{om1} = V_{CC}$

$P_{om} = \frac{V_{om}^2}{2R_L} = \frac{V_{CC}^2}{2R_L}$

$\therefore V_{CC} = \sqrt{2R_L P_{om}} = \sqrt{2 \times 8 \times 9} = 12V$

在OTL时, 由于只有1个电源, 则 $V_{CC}' = 2V_{CC} = 24V$

3.3 (1). 根据互补对称共集电路的对称性知:

$V_{C2} = \frac{1}{2} V_{CC} = 5V$

为了改变电容两端的电压差可以调节 R_1 , R_3 来改变电路对称性

(2). 有两方面来解释.

① 交越失真 是因输入信号幅度过大导致某点电压小于开启电压, 从而使 T_1, T_2 全截止.

$V_{e1} = V_{e2} = \frac{1}{2} V_{CC}$

$V_{b1} = \frac{R_2 + R_3}{R_1 + R_2 + R_3} V_{CC}, \quad V_{b2} = \frac{R_3}{R_2 + R_1 + R_3} V_{CC}$

不妨令 $R_1 = R_3 = R$.

$|V_{be1}| = \frac{R_2}{2(R_2 + 2R)} V_{CC}, \quad |V_{be2}| = \frac{R_2}{2(R_2 + 2R)} V_{CC}$

我们想尽可能提高 $|V_{be1}| \Rightarrow$ 减小电阻 R 或增大电阻 R_2 .

② 交越失真也可以从电流角度解释, 要提高 b_1, b_2 间电压, 提供较大的静态电流, 则应减小电阻 R 或提高 R_2 来增大 b_1, b_2 间电压.

(3). 开路后, 作静态分析

由于 $I_{B1} = I_{B2} \Rightarrow I_{B1} = I_{B2}, I_{C1} = I_{C2}$.

$I_{B1} = I_{B2} = \frac{V_{CC} - 2V_{BE}}{R_1 + R_3} = \frac{10 - 2 \times 0.7}{24} = 3.583mA$

$I_{C1} = I_{C2} = \beta I_{B1} = 179.167mA$

$V_{CE1} = V_{CE2} = \frac{1}{2} V_{CC} = 5V$

则 $P_C = I_{C1} \cdot V_{CE} = 895.833mW > P_{CM}$

$\therefore$ 功率管会烧坏.

3.4 (1). $V_{CEQ} = \frac{1}{2} V_{CC} = 15V$ (根据OTL的对称性)

$I_{CQ} = \frac{V_{CC} - V_{CEQ}}{R_C} = \frac{30 - 15}{1.5} = 10mA$

(2). 转化为相应的OCL电路分析.

$V_{CC}' = \frac{1}{2} V_{CC} = 15V$

最大饱和不失真 $V_{om1} = V_{CC}' - V_{CES} = 12V$

则 $V_{om} = 12V$

$P_{om} = \frac{V_{om}^2}{2R_L} = \frac{12^2}{2 \times 8} = 9W$

$P_E = \frac{2V_{om} V_{CC}}{\pi R_L} = \frac{2 \times 12 \times 15}{\pi \times 8} = 14.324W$

$\eta = \frac{P_{om}}{P_E} \times 100\% = \frac{\pi}{4} \frac{V_{om}}{V_{CC}} = \frac{\pi}{4} \times \frac{12}{15} = 62.83\%$

3.5 (1). 并联电压负反馈.

根据虚短虚断, $\frac{V_i - 0}{R_1} + \frac{V_o - 0}{R_2} = 0$

$\therefore A_{vf} = -\frac{R_2}{R_1} = -10$

(2). 最大饱和不失真 $V_{om1} = V_{CC} - V_{CES} = 15 - 0.7 = 14.3V$

最大输出电压 $V_{om2} = V_o - V_{BE} = 10 - 0.7 = 9.3V$

最大输出电流 $V_{om3} = R_L I_{Omax} = R_L \cdot (1 + \beta) I_{Bmax}$
 $= R_L \cdot (1 + \beta) I_m$

(由于未知 β , 最后一项可不作考虑)

$V_{om} = 9.3V$

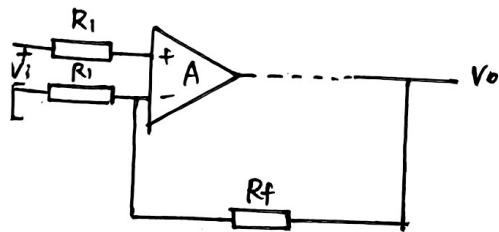
OCL中, $P_{om} = \frac{V_{om}^2}{2R_L} = \frac{9.3^2}{2 \times 8} = 5.406W$

(3). $P_E = \frac{2V_{om} V_{CC}}{\pi R_L} = \frac{2 \times 9.3 \times 15}{\pi \times 8} = 11.101W$

$P_T = \frac{1}{2} (P_E - P_{om}) = \frac{1}{2} \times (11.101 - 5.406) = 2.848W$

$\eta = \frac{P_{om}}{P_E} \times 100\% = \frac{\pi}{4} \frac{V_{om}}{V_{CC}} = \frac{\pi}{4} \times \frac{9.3}{15} = 48.695\%$

3.6 (1). 提高 $R_1 \rightarrow$ 串联 负反馈
稳定 $V_o \rightarrow$ 电压



(2). 由虚短, 虚断性质:

$$V_f = \frac{R_1}{R_1 + R_f} V_o, \quad V_f = V_i$$

$$\therefore A_{vf} = 1 + \frac{R_f}{R_1} = 50$$

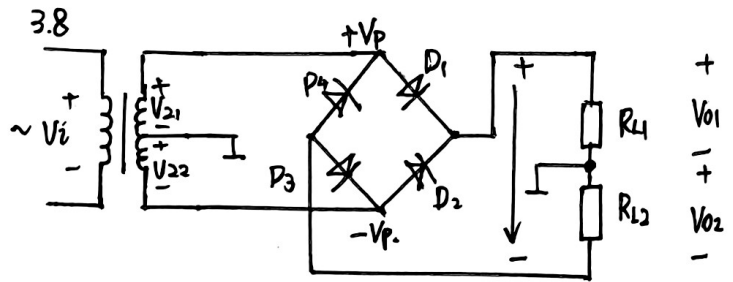
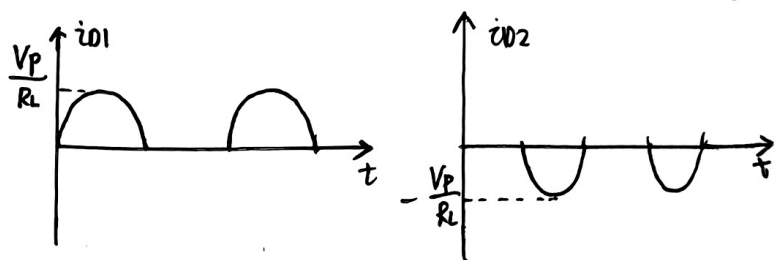
$$\Rightarrow R_f = 49R_1 = 49k\Omega$$

(3). 最大饱和不失真 $V_{om1} = V_{cc} - V_{CES} = 15 - 1 = 14V$

$$\text{最大输出电压 } V_{om2} = V_{omax} - V_{BE} + V_T = 13 - 0.7 + 0.7 = 13V$$

$$\text{则 } V_{om} = 13V$$

$$P_{om} = \frac{V_{om}^2}{2R_1} = \frac{13^2}{2 \times 8} = 10.563W$$



(1). V_{o1} 对地为 +, V_{o2} 对地为 -

$$(2). V_p = 20\sqrt{2} V.$$

$$V_{o1p} = V_{o2p} = 10\sqrt{2} V.$$

$$\text{平均值: } V_{o1(AV)} = -V_{o2(AV)} = \frac{2}{\pi} \int_0^\pi V_{o1p} \sin t dt = \frac{2V_{o1p}}{\pi} = \frac{20\sqrt{2}}{\pi} V \approx 20 \times 0.9 = 18V$$

$$\text{则 } V_{o1(AV)} = \frac{20\sqrt{2}}{\pi} V, \quad V_{o2(AV)} = -\frac{20\sqrt{2}}{\pi} V.$$

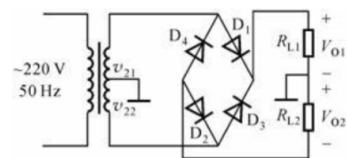
$$(3). \begin{matrix} V_{21}/V = 9V \\ V_{22}/V = 9V \end{matrix}$$

题3.8 图题3.8所示为桥式整流电路。

(1) 分别标出 V_{o1} 和 V_{o2} 对地的极性;

(2) 当 $V_{21}=V_{22}=20V$ (有效值) 时, 输出电压平均值 $V_{O(AV)1}$ 和 $V_{O(AV)2}$ 各为多少?

(3) 当 $V_{21}=18V, V_{22}=22V$ 时, 画出 V_{o1} 、 V_{o2} 的波形, 并求出 V_{o1} 和 V_{o2} 各为多少?



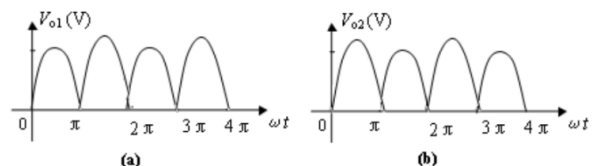
图题 3.8

解: (1) V_{o1} 上“+”下“-”, V_{o2} 上“+”下“-”

$$(2) V_{O(AV)1} = V_{O(AV)2} = 0.9V_{21} = 18(V)$$

$$(3) V_{O(AV)1} = V_{O(AV)2} = \frac{1}{2} \times 0.9(V_{21} + V_{22}) = \frac{0.9 \times (18 + 22)}{2} = 18(V), \quad V_{o1} \text{ 和 } V_{o2} \text{ 的波形}$$

分别如 (a) 和 (b) 所示。



(2). 考虑一个周期内: 设峰值为 V_p .

$$V_{21(AV)} = V_{22(AV)} = \frac{1}{2\pi} \int_0^{2\pi} V_p \sin t dt = 0$$

$$V_{O(AV)} = \frac{2}{\pi} \int_0^\pi V_p \sin t dt = \frac{2V_p}{\pi} = 0.9V_{21} = 0.9V_{22}$$

$$(3). I_{D(AV)1} = I_{D(AV)2} = \frac{1}{2\pi} \int_0^\pi \frac{V_p \sin t}{R_L} dt = \frac{V_p}{\pi R_L}$$

$$I_{L(AV)} = \frac{2}{2\pi} \int_0^\pi \frac{V_p \sin t}{R_L} dt = \frac{2V_p}{\pi R_L}$$

(因为电流的图像与电压不同, 所以结果不同)

(4). 前半周期, D_1 导通, D_2 截止.

$$V_o = V_{21},$$

$$V_{(BR)} = V_{op} + V_{22p} \Rightarrow V_{(BR)} \geq (\sqrt{2}V_{21} + \sqrt{2}V_{22}) \cdot \sqrt{2}$$

$$\text{后半周期同理} \quad = 2\sqrt{2} V_{21}$$

3.9 (1) 只有整流部分

$$V_{O(AV)} = \frac{2}{2\pi} \int_0^\pi V_p \sin t dt = \frac{2V_p}{\pi}$$

$$= \frac{40\sqrt{2}}{\pi} (V) = 18V = 0.9V_2$$

(2) 空载时,

$$V_{O(AV)} = \sqrt{2} V_2 = 20\sqrt{2} V$$

(3) 半波整流+滤波, 空载.

$$V_{O(AV)} = \sqrt{2} V_2 = 20\sqrt{2} V \text{ (电容器充满后不再变化)}$$

(4) 桥式整流+滤波.

$$V_{O(AV)} = 1.2 V_2 = 24V.$$

3.10 电阻R短路认为是桥式整流+滤波且有负载的电压 V_0'

$$V_0' = 1.2 V_2$$

$$V_{O(AV) \max} = 1.2 V_2 (1+10\%) = 23.76V$$

$$V_{O(AV) \min} = 1.2 V_2 (1-10\%) = 19.44V$$

$$(1) R_{\max} = \frac{V_{O(AV) \min} - V_D}{I_D + I_{L \max}} = \frac{19.44 - 5}{5 + 30} = 412.57 \Omega$$

$$(2) I_{Z \max} = \frac{V_{O(AV) \max} - V_D}{R} - I_{L \min}$$

$$= \frac{23.76 - 5}{0.41257 k} = 45.47 \text{ mA.}$$

思考: 为什么要取 $R = 370 \Omega$?

$$3.11 (1) \text{ 虚断性质, } I_Z = I_{R1} = \frac{V_I - V_Z}{R_1} = \frac{20 - 8}{1} = 12 \text{ mA.}$$

串联电压负反馈.

$$(2) V_+ = V_- = 8V$$

$$V_- = V_+ = 8V \text{ (虚短)} = V_f.$$

$$\text{反馈网络: } V_f = \frac{R_3}{R_2 + R_3} V_0$$

$$V_0 = \frac{R_2 + R_3}{R_2} V_f = (1 + \frac{R_3}{R_2}) V_f = 16V$$

$$(3) V_0 = (1 + \frac{R_3}{R_2}) V_Z$$

$$V_{O(\min)} = V_Z = 8V$$

$$\text{最大不失真幅度 } V_{Om1} = V_{CC} - V_{CES} = 20V$$

$$\text{输入电压限制 } V_{Om2} = (1 + \frac{R_{3 \max}}{R_2}) V_Z = 32V$$

$$\therefore V_{O(\max)} = 20V$$

3.12 (1) 稳压电路要求负反馈.

故A的上端应标+, 下端标-.

串联电压负反馈

$$(2) \text{ 最大饱和不失真 } V_{Om1} = V_{CC} = 24V$$

$$\text{输入电压限制: } V_{Om2} = V_0 - V_{BE}$$

$$= 20 - 0.7$$

$$= 19.3V$$

$$\text{输入电流限制: } I_{O \max} = (1+\beta) I_{O \max}$$

$$= 0.101 A, \approx 100 \text{ mA.}$$

$$\therefore \text{输出 } V_{Om} = 19.3V, I_{Om} = 100 \text{ mA.}$$

$$P_{CM} = I_{CM} \cdot V_{CEM}$$

$$= 100 \times (V_{CC} - V_{Om})$$

$$V_{O \min} \text{ 计算: } V_+ = V_- = V_D = V_f$$

$$V_f = \frac{R_w}{R_w} V_0$$

$$V_0 = \frac{R_w}{R_w} V_f \Rightarrow V_{O \min} = V_D = 6V$$

$$P_{CM} = 100 \times (24 - 6) = 1.8W$$

(1)

3.13 根据T型

$$V_B = V_Z + V_{BE} = 0.7 + 6.3 = 7V$$

V_B 和 V_0 之间的关系受 R_1, R_w, R_2 调节

$$\text{当接头在 } R_w \text{ 最上侧, } V_{O \min} = \frac{R_1 + R_w}{R_1 + R_2} V_B$$

$$\text{当接头在 } R_w \text{ 最下侧, } V_{O \max} = \frac{R_1 + R_2 + R_w}{R_2} V_B$$

$$\Rightarrow \begin{cases} R_w = 350 \Omega \\ R = 300 \Omega \end{cases}$$

(2) 记 R_4 上端电压为 V_0'

由于整流滤波作用, $V_0' = 1.2 V_2$

$$V_{CE1} = V_0' - V_0$$

$$\text{则 } V_{O' \min} = V_{CE1} + V_{O \max} = 4 + 20 = 24V$$

(保证 V_0 在 $10 \sim 20V$ 内均能正常工作, 故 $20V$ 也要满足)

$$\text{则 } V_{2 \min} = \frac{24}{1.2} = 20V$$

3.14. 根据稳压模块 $V_0' = V_{xx}$

$$V_f = V_0 - V_0' = V_0 - V_{xx}$$

$$V_- = \frac{R_4}{R_3 + R_4} V_f$$

$$V_+ = \frac{R_2}{R_1 + R_2} V_0$$

$$3 \text{ 虚短, } V_- = V_+ \Rightarrow \frac{R_4}{R_3 + R_4} (V_0 - V_0') = \frac{R_2}{R_1 + R_2} V_0$$

3.15 整流滤波后的等效 V_2 是:

$$V_2 = 40V \text{ (因为上端是 } 20\sqrt{2}\sin\omega t, \text{ 下端是 } -20\sqrt{2}\sin\omega t)$$

(有效值为 $40V$).

(1) C_1 : 上正下负; C_2 : 上正下负

W78xx 产生正电压.

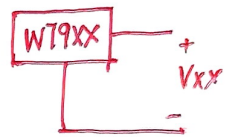
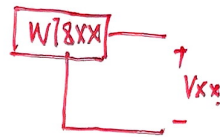
(2) 由于稳压模块的作用.

$$V_{R1} = 15V, V_{R2} = 15V$$

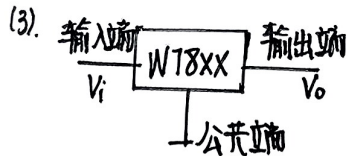
W79xx 产生负电压.

根据地在中间的极性, 有: $V_{01} = +15V, V_{02} = -15V$

表示实际上 V_{01} 上端为 +, V_{02} 下端为 +.



都以中端点引出点为 - 级.



$$V_i = 1.2 \times V_o \cdot \frac{1}{2} = 24V \text{ (因为中间是地, 两边电压被平分)}$$

$$V_o = 15V.$$

$$V_{W78xx} = V_i - V_o = 9V$$

$$I_{W78xx} = I_{L1} = 1A$$

$$P_{CM} = 9 \times 1 = 9W$$

3.16 (a). 稳压器反向工作时才能起稳压作用, 因此需要上下颠倒.

(b). 稳压器 W7805 应输入直流信号, 所以滤波电容要移至稳压器前.

(c). W7905 的 3 号引脚接错, 应接到 $-V_i$ 上.

修改方式: 把 V_i 分割为 $+\frac{V_i}{2}, -\frac{V_i}{2}$.

